I Maratona Padawan de Programação do IFB

October 17, 2026

Judging environment and submission system

This document describes the environment that compiles and runs the submissions of **I Maratona Padawan de Programação do IFB**. You can edit this text in the 📄 **Documents** tab of the contest panel. The system fills the marked blocks.

1. Judging environment

1.1 Environment

The judging system runs on **Ubuntu 24.04.4 LTS x86_64**. The judging machines report these compilers and interpreters:

- C: gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
- C++: g++ (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
- Java: javac 21.0.11 / javac 21.0.12
- Kotlin: info: kotlinc-jvm 2.1.20 (JRE 21.0.11+10-1-24.04.2-Ubuntu) / info: kotlinc-jvm 2.1.20 (JRE 21.0.12+8-1-24.04-Ubuntu)
- Python 3 (PyPy3): Python 3.9.18 (7.3.15+dfsg-1build3, Apr 01 2024, 03:12:48)

1.2 Accepted languages

The judge accepts these languages and file extensions:

Accepted languages	File extension
C	`.c`
C++	`.cpp, .cc, .cxx, .c++`
Java	`.java`
Kotlin	`.kt`

Accepted languages	File extension
Python 3 (PyPy3)	`.py`

The extension of the file that you submit identifies the language. Submit your source code with the correct extension. The judge rejects a file with an extension that is not in the table. To rename a file does not change the language: the server checks the language again when the submission arrives.

1.3 Memory limits

Every submission, in every language, has a total memory limit of **1024 MB** and a maximum stack size of **128 MB**. The judge measures the peak resident memory of the submission (the memory that the program touches), not the address space that it reserves.

Note: the memory limit includes the memory of the runtime. In Java, Kotlin and Python the runtime uses several megabytes. In C and C++ the C runtime is small, but it is not zero.

1.4 Time limits

Before the contest, the judges solve every problem. The time limit of each problem comes from the running time of those solutions. The current table is:

Problem	Name	C	C++	Java	Python
A	Alpinismo	—	0.34	1.06	1.3
B	Bodybuilder	0.04	0.04	0.84	0.14
C	Criptografia	—	5.36	3	7.94
D	Dança das Cadeiras	0.04	0.04	1.24	0.2
E	Estratégia de Pneus	0.04	0.04	1.62	0.14
F	Foos Contra Fighters	0.06	0.08	0.62	0.2

Problem	Name	C	C++	Java	Python
G	Guerra nas Estrelas	0.04	0.04	0.14	0.18
H	Hertz pra todos	0.18	0.18	2.06	1.22
I	Indo a Uberlândia	0.04	0.04	0.6	0.14
J	Juntando os Dígitos	0.04	0.06	0.6	0.2
K	Kernel	0.58	0.56	—	4.3
L	Leitura de Relatórios	0.04	0.04	1.78	0.12
M	Mentalista	0.04	0.04	0.46	0.18

* The time limit depends on the language: one column per language.

The time limit of each problem is also available in the MOJ interface.

1.5 Other limits

- Source file size: 1024 KB.
- Program output size, including the standard error output: 250 MB.
- Compilation time: 30 seconds.

1.6 Compilation and execution

Compile and run your solutions with the commands in the sections below. These commands do not apply the memory and time limits of the judge, but they match the environment of the judge.

C and C++

- The judge runs the executable file that the compiler produces.
- Your program must exit with status zero: end `main` with `return 0`, or call `exit(0)`.
- For problems with large inputs, `cin` and `cout` can be slow. By default they use a buffer synchronized with the `stdio` library. To use `cin` and `cout`, call

`std::ios::sync_with_stdio(false);` at the start of `main`. In this case, do not mix `scanf` and `printf` in the same program.

- Compile C solutions with:

```
gcc -lm -O2 -static {submitted_file} -o main
```

- Compile C++ solutions with:

```
g++ -lm -O2 -static -std=gnu++20 -pipe {submitted_file} -o main
```

- Run C and C++ solutions with:

```
./main
```

Java

- The judge runs the compiled class that contains `main`.
- Do not declare a `package` in your program. A program with a package does not run in MOJ.
- The name of the file must match the public class. If your file is `Klass.java`, the public class with `main` must be `Klass`. Otherwise you can get a Compilation Error.
- For problems with large inputs, `Scanner` can be slow. Use buffered I/O, for example `BufferedReader` and `PrintWriter`.
- Compile Java solutions with:

```
javac -encoding UTF-8 {submitted_file}
```

- Run Java solutions with:

```
java -Duser.language=en -Duser.country=US \
     -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8 \
     -Xms10m -Xmx1024m -Xss131072k {class_name}
```

- The judge sets the *locale* to `en_US` and the output to UTF-8: `Scanner.nextDouble()` reads `5.5` and `printf("%.2f")` writes a dot.

- The judge sets the heap and stack sizes above from the memory limits of section 1.3. #### Kotlin

- The judge runs the compiled program.

- Do not declare a `package` in your program. A program with a package does not run in MOJ.

- There is no rule for the file name. The entry point is the top-level `fun main()` of the file that you submit.

- Compile Kotlin solutions with:

```
kotlinc {submitted_file} -include-runtime -d prog.jar
```

- Run Kotlin solutions with:

```
java -Duser.language=en -Duser.country=US \
     -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8 \
     -Xms10m -Xmx1024m -Xss131072k -jar prog.jar
```

- The judge sets the *locale* to `en_US` and the output to UTF-8, as in Java.

- The judge sets the heap and stack sizes above from the memory limits of section 1.3. #### Python

- The judge runs the file that you submit with the **PyPy3** interpreter, not with CPython. PyPy3 is usually faster, but some behaviors are different.

- The judge checks the syntax of the file when you submit it. A file with a syntax error gets a Compilation Error.

- Check the syntax of Python solutions with:

```
pypy3 -m py_compile {submitted_file}
```

- Run Python solutions with:

```
pypy3 {submitted_file}
```

For other accepted languages, the judge compiles and runs the file with the standard tools of

the language, with the same limits.

2. How to use the MOJ system

2.1 Verdicts

You can receive these verdicts:

- Not Answered Yet
- Accepted
- Wrong Answer
- Time Limit Exceeded
- Memory Limit Exceeded
- Runtime Error
- Compilation Error

In a contest the verdict is all that you receive: no failed test case, no diff and no partial score. The test data is contest material.

If you think that a verdict is wrong, use the **Clarification** tab. Identify the submission by problem and time.

2.2 Judging notes

- The judge can run your program on several input files. If your program has more than one error, for example Time Limit Exceeded and Wrong Answer, you can receive any of them.
- If the compilation takes too long, or if the compiled program is too large, you receive a Compilation Error.
- There is no Presentation Error and no Format Error. If the problem asks for the word `impossible` and you write it with a typo, the verdict is Wrong Answer.
- Follow the output format of the sample output. Extra whitespace within reason is accepted: a space at the end of a line, a space between tokens, or a blank line at the end. Thousands of blank spaces are not accepted.
- For problems with floating-point output, the judge accepts any answer inside the tolerance of the statement. The tolerance is absolute or relative, as the statement says. Your answer does not need to match the digits of the sample.
- When the statement says that several outputs are acceptable, the judge checks the

format and the constraints. The judge does not compare your output with the sample output.

- If your program writes too much output, the verdict is Runtime Error. The standard error output counts. Remove debug messages before you submit.

2.3 Penalties

Each submission that receives a verdict other than Accepted, before the first Accepted verdict of the same problem, adds a penalty of **20 minutes** to the total time of the team. There is no penalty for problems that the team does not solve.

Verdicts without penalty: **Compilation Error**.

2.4 Response times

The time to get a verdict depends on the problem, on the verdict and on the moment of the contest. The judge runs submissions in parallel, so you can receive verdicts out of order. Some submissions need a manual check by the judges. In that case, a delay of some minutes is normal.

2.5 Input and output

1. Read the input from the **standard input**.
2. Write the output to the **standard output**.
3. The input has one test case and no extra data.
4. When a line has several values, one space separates them.

For more questions about the judging system, read the competitor documentation in the web interface.